

SIMATS ENGINEERING

ASSESSMENT TOOL 2: Logic Reasoning & Mapping Exercise

COURSE NAME: ARTIFICIAL INTELLIGENCE COURSE CODE: MLA01
AND EXPERT SYSTEMS

COURSE OUTCOME COVERED

CO3: Implement propositional and first-order logic using resolution, unification, and inference techniques for knowledge representation and reasoning. (BTL 3)

Assessment Tool Weightage: 15%

Total Marks: 100

Time: 60 Mins

No. of Questions: 5

Annexure A

Logic Reasoning & Mapping Exercise

Code of Conduct:

I T Bhupal/192425243 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: *bhupalreddy*

Criterion	Weightage	Marks Obtained
1. Problem Analysis and Knowledge Identification	15%	
2. Logical Representation and Mapping	20%	
3. Predicates, Variables, Constants and Quantifiers	15%	
4. Logical Connectives and Formula Construction	10%	
5. Inference and Logical Reasoning	15%	
6. Unification and Substitution	10%	
7. Resolution and Proof Construction	10%	
8. Mapping Diagram / Clarity and Presentation	5%	
Total	100%	

ANSWER ANY ONE OF THE FOLLOWING

Q.No	Use Case Scenario	
1	Smart Healthcare Diagnosis	A hospital uses an AI system to identify possible diseases from patient symptoms. Ravi has fever, cough, and body pain. Patients with fever and cough may have flu. Patients with flu and body pain may have viral infection. Patients with viral infection require medical attention. Tasks: (i) Identify facts, entities, predicates, variables, and constants. (ii) Represent the knowledge using Propositional Logic. (iii) Represent the knowledge using FOL with suitable quantifiers. (iv) Perform unification for two relevant predicates. (v) Apply resolution to prove whether Ravi has a viral infection. (vi) Show the complete inference sequence and final conclusion.
2	Student Academic Advising	A university AI system recommends academic support based on attendance and examination performance. Priya has attendance below 75% and has failed two courses. Students with low attendance require attendance counseling. Students who fail two or more courses require academic counseling. Students requiring both types of counseling need academic support. Tasks: (i) Identify the knowledge components. (ii) Convert the statements into Propositional Logic. (iii) Develop FOL representations using predicates and quantifiers. (iv) Identify suitable substitutions and perform unification. (v) Use resolution to prove whether Priya requires academic support. (vi) Explain the reasoning steps leading to the conclusion.
3	Intelligent Loan Approval	A bank wants to use an AI knowledge-based system for preliminary loan approval. Arun has stable employment, sufficient income, a high credit score, and good repayment history. Customers with stable employment and sufficient income qualify for preliminary approval. Customers with a high credit score and preliminary approval are classified as low-risk. Low-risk customers with good repayment history can be recommended for loan approval. Tasks: (i) Identify entities, facts, predicates, and rules. (ii) Map the scenario into Propositional Logic. (iii) Represent the rules using FOL.

		(iv) Demonstrate unification and substitution. (v) Use resolution to prove or disprove LoanApproved(Arun). (vi) Provide the final inference with justification.
4	Vehicle Fault Diagnosis	An automobile service center uses an expert system to diagnose vehicle faults. Car101 does not start and its headlights are dim. Vehicles that do not start and have dim headlights may have a battery problem. Vehicles with a battery problem require battery inspection. Vehicles requiring battery inspection are marked for service. Tasks: (i) Extract the facts and rules from the scenario. (ii) Represent them using Propositional Logic. (iii) Develop suitable FOL predicates and rules. (iv) Perform unification for the vehicle-related predicates. (v) Apply resolution to determine whether Car101 has a battery problem. (vi) Derive whether Car101 should be marked for service and explain the inference.
5	Cybersecurity Threat Detection	An organization uses an AI system to identify cyber threats. User101 has five failed login attempts and logged in from an unusual location. Multiple failed login attempts indicate a possible brute-force attack. A brute-force attack combined with an unusual login location indicates suspicious activity. Suspicious activity combined with unauthorized access to confidential files indicates a possible security breach. Tasks: (i) Identify the facts, entities, predicates, and rules. (ii) Map the knowledge into Propositional Logic. (iii) Represent the rules using FOL. (iv) Demonstrate unification and substitution. (v) Apply resolution to determine whether User101 is involved in a possible security breach. (vi) Present the complete reasoning chain and final inference.

Smart Healthcare Diagnosis

(i) Identify facts, entities, predicates, variables, and constants

Scenario Analysis: The scenario describes a medical diagnostic process mapping patient symptoms to diseases and subsequent actions.

- Facts (Symptoms): Ravi has a fever, Ravi has a cough, Ravi has body pain.
- Entities / Constants: Ravi (A specific patient).
- Variables: x, y, z (Representing any patient/person).
- Predicates:
 - Fever(x): x has a fever.
 - Cough(x): x has a cough.
 - BodyPain(x): x has body pain.
 - Flu(x): x has the flu.
 - ViralInfection(x): x has a viral infection.
 - RequiresMedicalAttention(x): x requires medical attention.

(ii) Represent the knowledge using Propositional Logic

Note: Propositional logic uses specific propositions for Ravi since it cannot handle variables.

Propositions:

- P: Ravi has a fever.
- Q: Ravi has a cough.
- R: Ravi has body pain.
- S: Ravi has the flu.
- T: Ravi has a viral infection.
- U: Ravi requires medical attention.

Knowledge Base (KB):

1. P (Fact)
2. Q (Fact)
3. R (Fact)
4. $P \wedge Q \rightarrow S$ (Rule 1)
5. $S \wedge R \rightarrow T$ (Rule 2)
6. $T \rightarrow U$ (Rule 3)

(iii) Represent the knowledge using FOL with suitable quantifiers

First-Order Logic introduces universal quantifiers ($\forall$) and variables to generalize the rules for any patient.

Facts:

1. $\text{Fever}(\text{Ravi})$
2. $\text{Cough}(\text{Ravi})$
3. $\text{BodyPain}(\text{Ravi})$

Rules (with Universal Quantifiers): 4. $\forall x (\text{Fever}(x) \wedge \text{Cough}(x) \rightarrow \text{Flu}(x))$ 5. $\forall y (\text{Flu}(y) \wedge \text{BodyPain}(y) \rightarrow \text{ViralInfection}(y))$ 6. $\forall z (\text{ViralInfection}(z) \rightarrow \text{RequiresMedicalAttention}(z))$

(iv) Perform unification for two relevant predicates

Unification is the process of making two different logical expressions identical by finding a substitution for their variables.

Target Predicates to Unify:

- The premise from Rule 4: $\text{Fever}(x)$
- The known fact: $\text{Fever}(\text{Ravi})$

Unification Step: To match $\text{Fever}(x)$ with $\text{Fever}(\text{Ravi})$, we apply a substitution where the variable x is replaced by the constant Ravi .

- Most General Unifier (MGU), $\theta: \{x / \text{Ravi}\}$
- Result after substitution: $\text{Fever}(\text{Ravi})$
- *(This substitution is uniformly applied to the rest of the rule, making it: $\text{Fever}(\text{Ravi}) \wedge \text{Cough}(\text{Ravi}) \rightarrow \text{Flu}(\text{Ravi})$)*

(v) Apply resolution to prove whether Ravi has a viral infection

To prove $\text{ViralInfection}(\text{Ravi})$ using resolution, we must use Proof by Contradiction. We negate the goal, convert the Knowledge Base to Conjunctive Normal Form (CNF), and resolve clauses until we reach an empty clause (contradiction).

Step 1: Convert KB to CNF (Clause Form) *Recall: $A \rightarrow B$ becomes $\neg A \vee B$. $\neg(A \wedge B)$ becomes $\neg A \vee \neg B$.*

1. $\text{Fever}(\text{Ravi})$
2. $\text{Cough}(\text{Ravi})$
3. $\text{BodyPain}(\text{Ravi})$
4. $\neg \text{Fever}(x) \vee \neg \text{Cough}(x) \vee \text{Flu}(x)$
5. $\neg \text{Flu}(y) \vee \neg \text{BodyPain}(y) \vee \text{ViralInfection}(y)$
6. $\neg \text{ViralInfection}(z) \vee \text{RequiresMedicalAttention}(z)$

Step 2: Negate the Goal 7. $\neg \text{ViralInfection}(\text{Ravi})$

Step 3: Resolution Steps

- Resolve (7) and (5) using substitution $\{y / \text{Ravi}\}$:
 - $\neg \text{ViralInfection}(\text{Ravi})$ resolves with $\neg \text{Flu}(\text{Ravi}) \vee \neg \text{BodyPain}(\text{Ravi}) \vee \text{ViralInfection}(\text{Ravi})$
 - *Derived Clause 8:* $\neg \text{Flu}(\text{Ravi}) \vee \neg \text{BodyPain}(\text{Ravi})$
- Resolve (8) and (3):
 - $\neg \text{Flu}(\text{Ravi}) \vee \neg \text{BodyPain}(\text{Ravi})$ resolves with $\text{BodyPain}(\text{Ravi})$
 - *Derived Clause 9:* $\neg \text{Flu}(\text{Ravi})$

- Resolve (9) and (4) using substitution $\{x / \text{Ravi}\}$:
 - $\neg \text{Flu}(\text{Ravi})$ resolves with $\neg \text{Fever}(\text{Ravi}) \vee \neg \text{Cough}(\text{Ravi}) \vee \text{Flu}(\text{Ravi})$
 - *Derived Clause 10:* $\neg \text{Fever}(\text{Ravi}) \vee \neg \text{Cough}(\text{Ravi})$
- Resolve (10) and (2):
 - $\neg \text{Fever}(\text{Ravi}) \vee \neg \text{Cough}(\text{Ravi})$ resolves with $\text{Cough}(\text{Ravi})$
 - *Derived Clause 11:* $\neg \text{Fever}(\text{Ravi})$
- Resolve (11) and (1):
 - $\neg \text{Fever}(\text{Ravi})$ resolves with $\text{Fever}(\text{Ravi})$
 - *Derived Clause 12:* $[\]$ (Empty Clause / Contradiction)

Since we reached a contradiction (Empty Clause), the negated goal is FALSE. Therefore, the original goal $\text{ViralInfection}(\text{Ravi})$ is TRUE.

(vi) Show the complete inference sequence and final conclusion

Forward Chaining Logical Sequence (Modus Ponens):

1. We know $\text{Fever}(\text{Ravi})$ and $\text{Cough}(\text{Ravi})$ are True (Given Facts).
2. By Universal Instantiation on Rule 1: $\text{Fever}(\text{Ravi}) \wedge \text{Cough}(\text{Ravi}) \rightarrow \text{Flu}(\text{Ravi})$.
3. Applying Modus Ponens to Step 1 & 2, we deduce: $\text{Flu}(\text{Ravi})$ is True.
4. We know $\text{BodyPain}(\text{Ravi})$ is True (Given Fact).
5. By Universal Instantiation on Rule 2: $\text{Flu}(\text{Ravi}) \wedge \text{BodyPain}(\text{Ravi}) \rightarrow \text{ViralInfection}(\text{Ravi})$.
6. Applying Modus Ponens to Step 3 & 4, we deduce: $\text{ViralInfection}(\text{Ravi})$ is True.

Final Conclusion: Based on the provided facts and logical rules, the system successfully infers and proves that Ravi has a viral infection and will require medical attention.



Smart Healthcare Diagnosis System

1. Input Patient Symptoms

- ☐ Patient has Fever
- ☐ Patient has Cough
- ☐ Patient has Body Pain

2. Forward Chaining Logic Engine

```
| 📄 FACTS IDENTIFIED:  
| └─ None  
  
| ⚙️ EVALUATING RULE 1: IF Fever AND Cough THEN Flu  
| └─ Conditions not met.  
  
| ⚙️ EVALUATING RULE 2: IF Flu AND BodyPain THEN Viral Infection  
| └─ Conditions not met.  
  
| ⚙️ EVALUATING RULE 3: IF Viral Infection THEN Medical Attention  
| └─ Conditions not met.
```

Awaiting symptoms for diagnosis...



Smart Healthcare Diagnosis System

1. Input Patient Symptoms

- ☒ Patient has Fever
- ☐ Patient has Cough
- ☐ Patient has Body Pain

2. Forward Chaining Logic Engine

```
| 📄 FACTS IDENTIFIED:  
| └─ Fever(Patient) = True  
  
| ⚙️ EVALUATING RULE 1: IF Fever AND Cough THEN Flu  
| └─ Conditions not met.  
  
| ⚙️ EVALUATING RULE 2: IF Flu AND BodyPain THEN Viral Infection  
| └─ Conditions not met.  
  
| ⚙️ EVALUATING RULE 3: IF Viral Infection THEN Medical Attention  
| └─ Conditions not met.
```

Symptoms present, but insufficient data to confirm Flu or Viral Infection.



Smart Healthcare Diagnosis System

1. Input Patient Symptoms

☒ Patient has Fever

☒ Patient has Cough

☒ Patient has Body Pain

2. Forward Chaining Logic Engine

FACTS IDENTIFIED:

└ Fever(Patient) = True

└ Cough(Patient) = True

└ BodyPain(Patient) = True

⚙️ EVALUATING RULE 1: IF Fever AND Cough THEN Flu

└ Modus Ponens Success: Flu(Patient) is TRUE

⚙️ EVALUATING RULE 2: IF Flu AND BodyPain THEN Viral Infection

└ Modus Ponens Success: ViralInfection(Patient) is TRUE

⚙️ EVALUATING RULE 3: IF Viral Infection THEN Medical Attention

└ Modus Ponens Success: RequiresMedicalAttention(Patient) is TRUE

 **TARGET PROVED: Patient has a Viral Infection and requires immediate medical attention.**

EVALUATION RUBRICS:

Criterion	Weightage (%)	Excellent	Good	Average	Poor
Problem Understanding & Knowledge Identification	15%	13–15% – Clearly identifies entities, facts, relationships, assumptions, and reasoning requirements from the given scenario.	10–12% – Identifies most relevant knowledge with minor omissions.	7–9% – Identifies basic knowledge but misses several important elements.	0–6% – Poor or incorrect identification of the given knowledge.
Logical Representation & Mapping	20%	17–20% – Accurately maps real-world statements into appropriate propositional and first-order logic representations.	13–16% – Provides mostly correct logical mappings with minor errors.	9–12% – Provides partial mappings with several logical errors.	0–8% – Logical mapping is largely incorrect or missing.
Predicates, Variables, Constants & Quantifiers	15%	13–15% – Correctly identifies and applies predicates, variables, constants, universal and existential quantifiers.	10–12% – Uses the concepts correctly with minor errors.	7–9% – Demonstrates partial understanding with some incorrect usage.	0–6% – Concepts are incorrectly applied or missing.
Logical Connectives & Formula Construction	10%	9–10% – Correctly applies AND, OR, NOT, implication, and equivalence to	7–8% – Mostly correct use of connectives with minor errors.	5–6% – Basic use of connectives but several errors occur.	0–4% – Incorrect or incomplete formula construction.

		construct valid logical formulas.			
Inference & Logical Reasoning	15%	13–15% – Performs accurate step-by-step reasoning and derives valid conclusions from the given knowledge.	10–12% – Reasoning is mostly correct with minor gaps.	7–9% – Provides partial reasoning with some incorrect conclusions.	0–6% – Reasoning is incorrect, unsupported, or missing.
Unification & Substitution	10%	9–10% – Correctly identifies substitutions and Most General Unifier (MGU) and applies them accurately.	7–8% – Correctly performs most unification steps with minor errors.	5–6% – Demonstrates basic understanding but has several errors.	0–4% – Unable to perform unification correctly.
Resolution / Proof Construction	10%	9–10% – Correctly applies resolution and presents a clear, logically valid proof of the query.	7–8% – Resolution is mostly correct with minor errors.	5–6% – Shows partial understanding but proof contains gaps.	0–4% – Resolution is incorrect or not demonstrated.
Clarity, Mapping Diagram & Presentation	5%	5% – Provides a clear, well-organized mapping/logic diagram with accurate notation and excellent presentation.	4% – Presentation is clear with minor formatting or diagram issues.	3% – Basic presentation with limited clarity.	0–2% – Poorly organized, unclear, or incomplete presentation.